Loan Risk ML Pipeline

End-to-End Credit Risk Modeling and Governance Workflow

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Business Problem –

Better Routing, Not Just Better Scores

01

Traditional statistical scoring :

Experts manually assign scores and rules to each attribute

Interpretable, but it depends heavily on expert judgment and manual maintenance

02

Typical ML approach :

Use historical data to improve prediction efficiency and reduce manual effort

It helps reduce manual review effort and supports more scalable prediction.

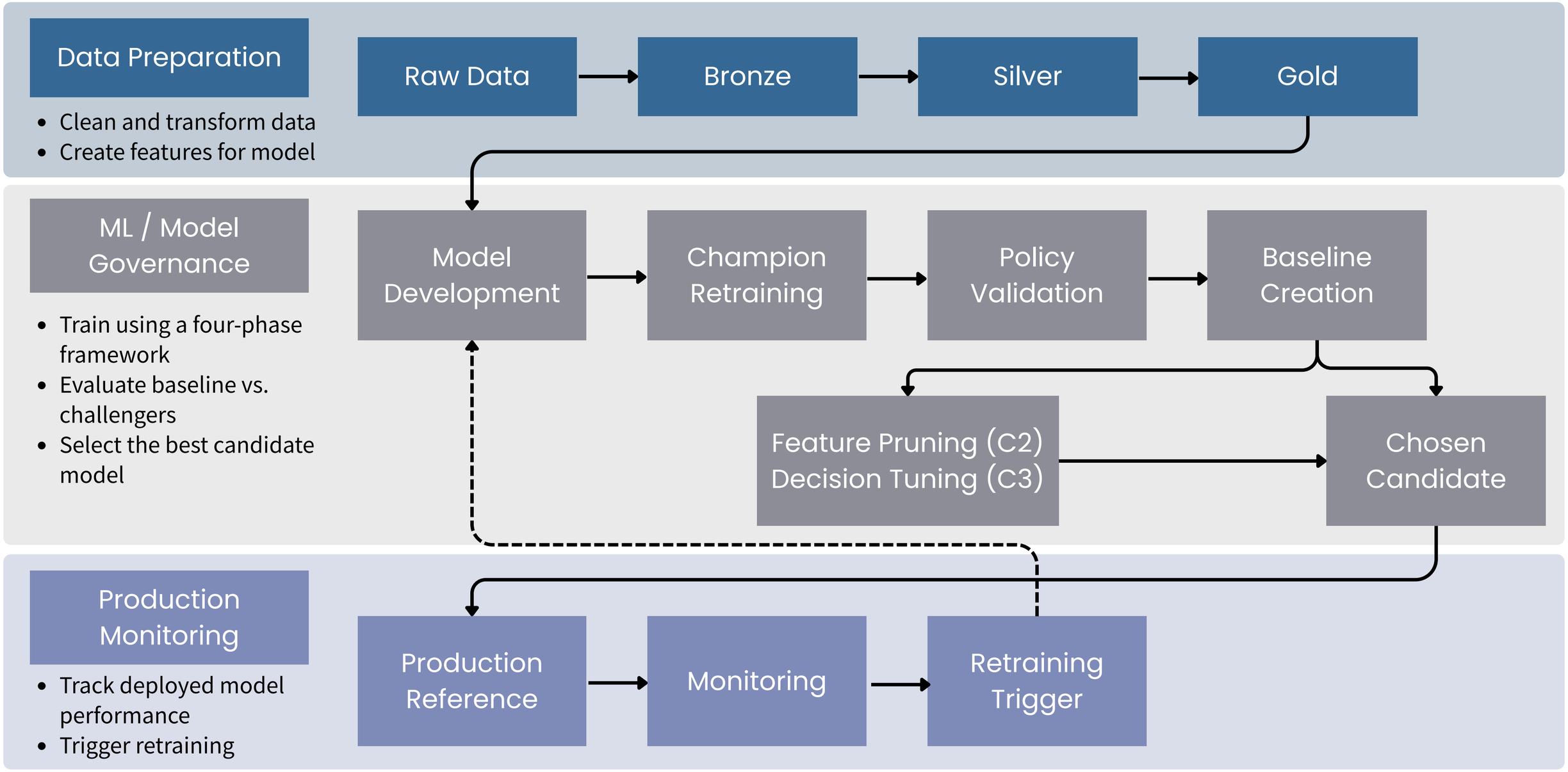
03

Our routing-first design :

Focus on routing cases into high, manual, and low review zones

Low Zone	Manual Zone	High Zone

End-to-End ML Workflow



Beyond Model Accuracy

The Need for Better Decision Routing

Problem

- Highly imbalanced data

~95% approvals, very few rejects

- Misleading aggregate metrics

High AUC may still miss risky cases

- One-size-fits-all decision

All cases treated the same way

Implication

- Implication
- Better review prioritization
- Better use of human effort

High accuracy does not guarantee useful decisions in an imbalanced setting

From Metrics to Decision Quality

Routing-first objective

The main objective of this project is not simply to maximize a classification score.
Instead, we prioritize the quality of routing across the review zones:

- High Zone
- Manual Zone
- Low Zone

Operational metrics

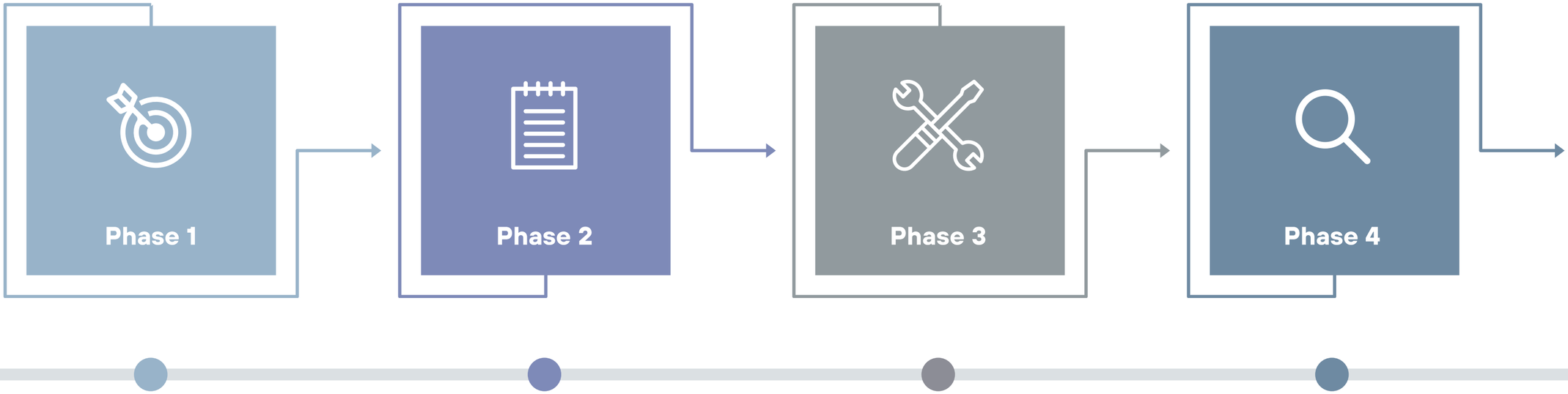
We focus on whether the model supports a more useful decision structure, especially through:

- high-zone approve precision
- low-zone reject precision
- human review workload ratio

Metrics we use

- AUC
- F1

Training Framework



Model Development

- Use **18 months** of historical data for **rolling training and monitoring**
- This phase is used to compare candidate strategies and assess how stable they are over time

Champion Retraining

After identifying the best strategy in Phase 1, we retrain it on the full development dataset to create the formal champion model

Policy Validation

- Use the following **4 months** to **tune thresholds** and routing policies.
- This phase focuses on how the model is used for operational decision routing.

Final Blind Holdout

- The **last 2 months** are kept completely untouched until the very end.
- This allows us to evaluate the final chosen strategy on unseen data.

Baseline and Challenger Design

Baseline

The first successful four-phase run becomes the official baseline and serves as the active production reference.
This baseline is the model against which future alternatives are compared.

C2 — Feature Pruning

It removes unstable or low-importance features and retrains the model to test whether a leaner design can improve stability and routing quality

C3 — Decision Tuning

Rather than retraining a completely new model, it focuses on optimizing the decision layer, such as thresholds and routing policy, to improve practical case routing

Final Model Decision

Model selection is based on:

- operational usefulness
- routing quality
- acceptable model guardrails

1

Retain Baseline

If neither challenger provides meaningful improvement, the baseline remains the official production reference

2

Full Upgrade

If a challenger improves routing quality while also satisfying classifier guardrails, it can be considered a full upgrade candidate

3

Routing-Only Upgrade

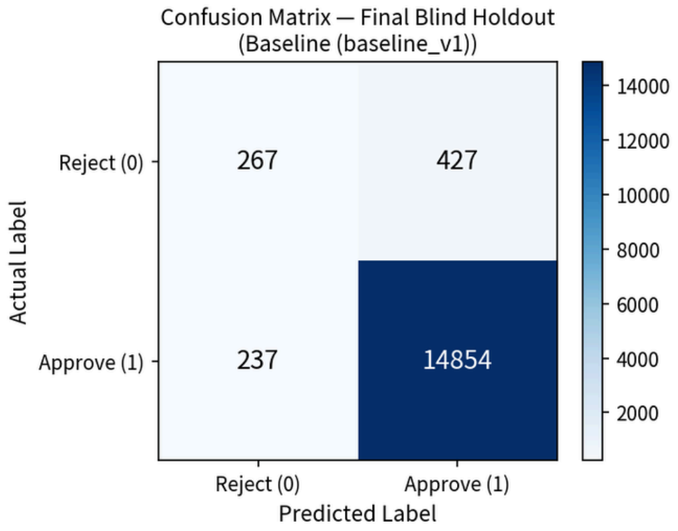
If a challenger improves routing outcomes but does not justify replacing the model itself. It improves routing policy, but does not justify replacing the model

Final Decision Report

Performance summary

Show final evaluation results on the Final Blind Holdout, including:

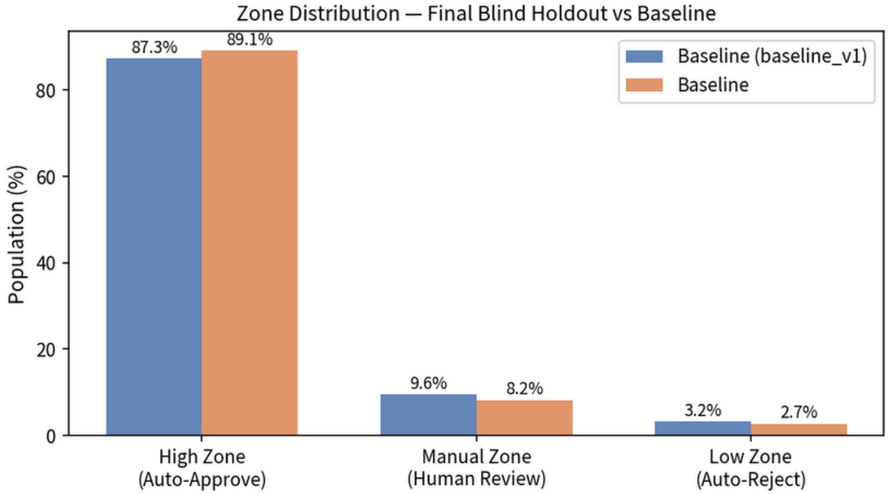
- **AUC**
- **F1**
- **confusion matrix**



Routing evidence

Help users understand how cases are distributed across:

- **High Zone**
- **Manual Zone**
- **Low Zone**

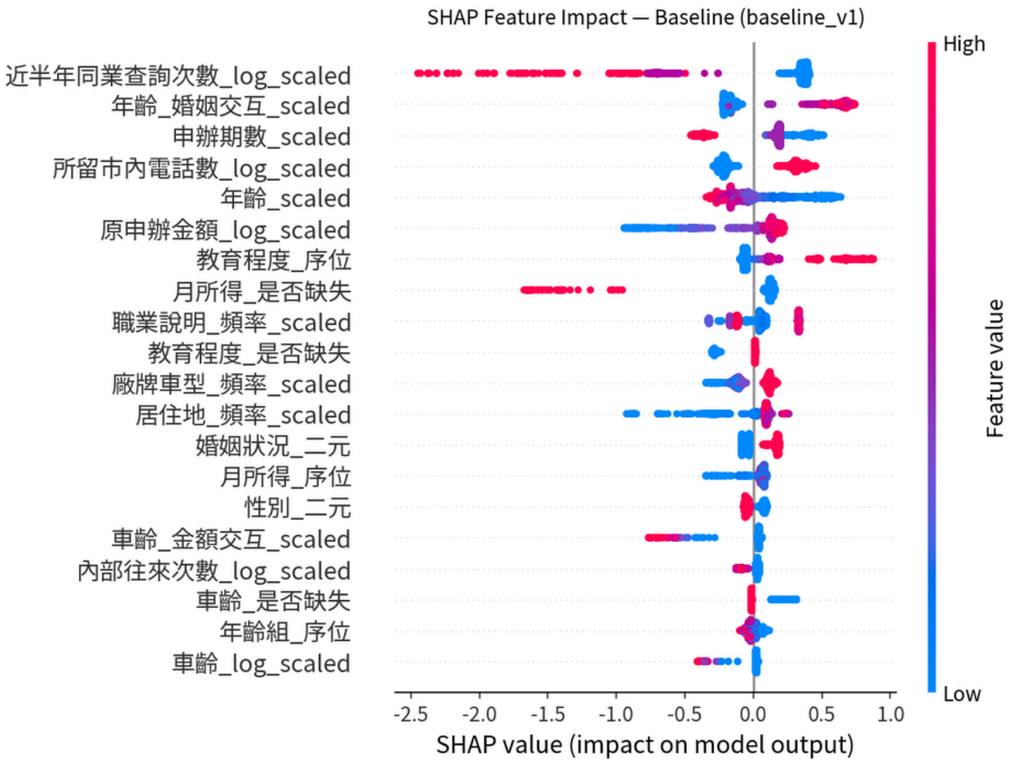


Final Decision Report

Explainability and visual support

Provide SHAP plots and other visual evidence to help users understand:

- **which features drive the model**
- **why the final model or strategy was retained**
- **how the decision outcome can be interpreted more clearly**



Monitoring and Model Stability

Performance tracking

After a production reference is chosen, we continue to monitor whether its predictive quality remains acceptable over time.

Drift detection

We also track whether the data or prediction distributions are shifting.
This includes monitoring using PSI on features and predictions.

Retraining triggers

Retraining is triggered under two conditions:

- Performance degradation

When model performance (e.g. AUC / F1) drops below acceptable levels

- Data drift detection

When PSI indicates significant shifts in feature or prediction distributions

Thank You